

RAHUL R – 192412255

ECA0502 EMF

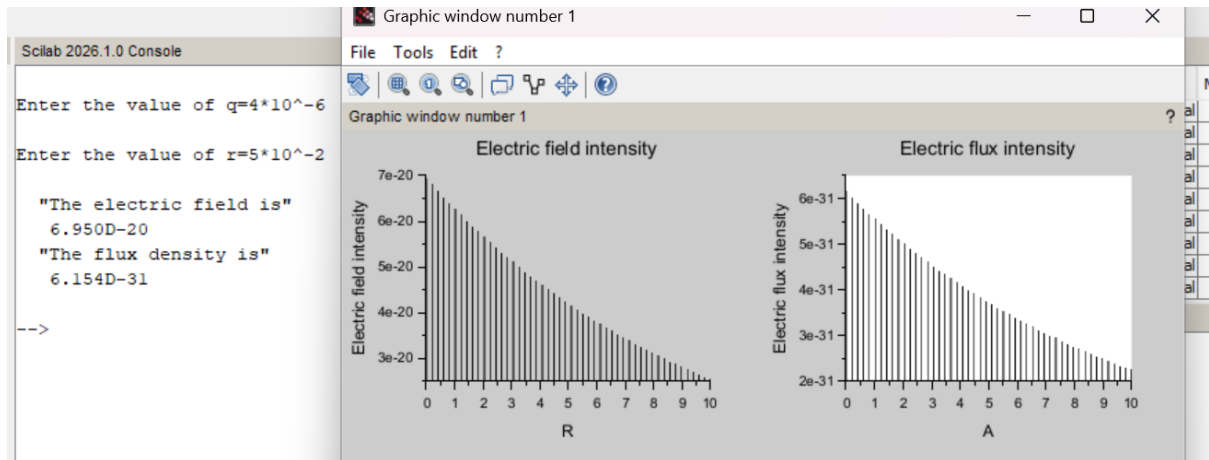
EXP 2

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

PROGRAM:

```
1  clc;
2  clear;
3
4  q=input('Enter the value of q=');
5  r=input('Enter the value of r=');
6
7  efcilon=8.854*10^-12;
8  pi=3.14
9  E=q/4*pi*efcilon*r^2;
10 disp('The electric field is', [E]);
11 D=E*efcilon;
12 disp('The flux density is', [D]);
13 x= linspace(0, 10, 50);
14 y=E*exp(-0.1*x);
15 z=D*exp(-0.1*x);
16
17 figure();
18 subplot(2, 2, 1);
19 plot2d3(x,y);
20 xlabel('R');
21 ylabel('Electric field intensity');
22 title('Electric field intensity');
23 subplot(2, 2, 2);
24 plot2d3(x,z);
25 xlabel('A');
26 ylabel('Electric flux intensity');
27 title('Electric flux intensity');
28
```

OUTPUT:



OBSERVATION:

EXP-2 Electric field & Electric flux density

i) $q = 4 \times 10^{-6} \text{ C}$

ii) $r = 5 \text{ cm}$

$E_0 = 8.854 \times 10^{-12} \text{ F/m}$

$E = \frac{q}{4\pi\epsilon_0 r^2}$

$E = \frac{4 \times 10^{-6}}{2.7799 \times 10^{-13}}$

$E = 6.950 \times 10^{-20} \text{ V/m}$

$D = \epsilon_0 E$

$D = 8.854 \times 10^{-12} \times 6.950 \times 10^{-20}$

$D = 6.154 \times 10^{-31}$

o/p ✓

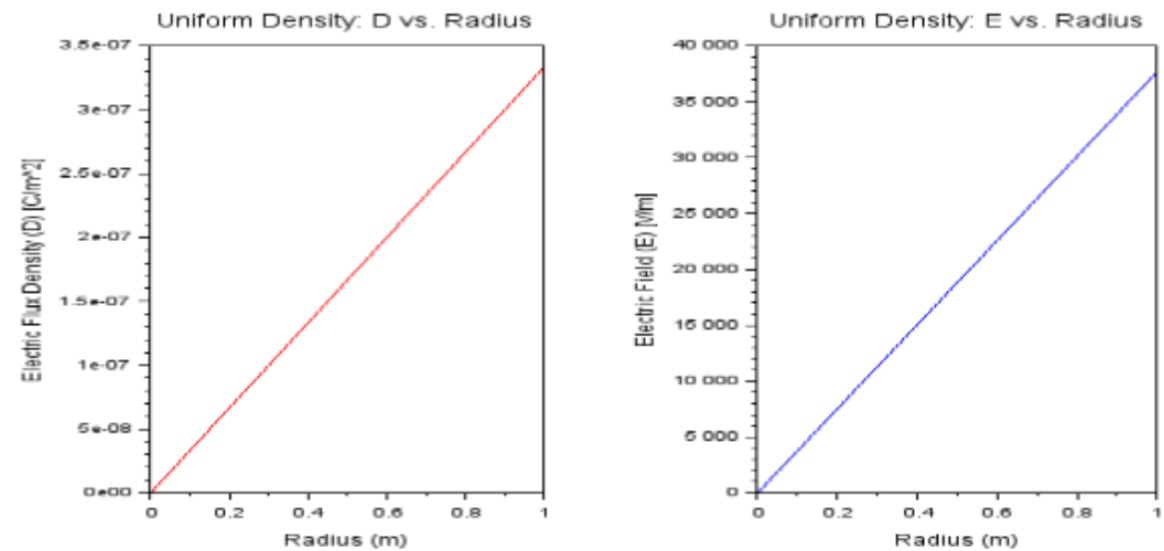
CASE 1:

PROGRAM:

```
EMF EX2 (II).sci (C:\Users\rahu\OneDrive\Documents\EMF EX2 (II).sci) - SciNotes
File Edit Format Options Window Execute ?
[Icons]
EMF EX2 (II).sci (C:\Users\rahu\OneDrive\Documents\EMF EX2 (II).sci) - SciNotes
EMF EX2 (I).sd X EMF EX2 (II).sd X EMF EX2 CASE 2.sd X EMF EXP2 CASE 3.sd X

1 rho0=1e-6; //Uniform charge density (C/m^3)
2 epsilon = 8.854e-12; // Permittivity of free space (F/m)
3 R=1; // Radius of the sphere (m)
4 n = 100; // Number of steps for plotting
5 r = linspace(0, R, n); // Radial distances
6
7 // Initialize arrays for D and E
8 D= zeros(1, n);
9 E= zeros(1, n);
10 // Calculate D and E
11 for i= 1:n
12   ri = r(i); // Current radius
13   // Enclosed charge Q_enclosed = rho0 * (4/3) * pi * r^3
14   Q_enclosed = rho0 * (4/3) * pi * ri^3;
15
16   // Electric Flux Density D
17   if ri > 0 then
18     D(i)=Q_enclosed/(4*pi*ri^2);
19   else
20     D(i) = 0; // Avoid division by zero
21   end
22
23   // Electric Field E = D ./ epsilon
24   E(i)= D(i) /epsilon;
25 end
26
27 // Plotting
28 subplot(1,2,1);
29 plot(r, D,"r");
30 xlabel("Radius (m)");
31 ylabel("Electric Flux Density (D) - [C/m^2]");
32 title("Uniform Density: D vs. Radius");
33
34 subplot(1, 2, 2);
Line 10, Column 11.
```

OUTPUT:



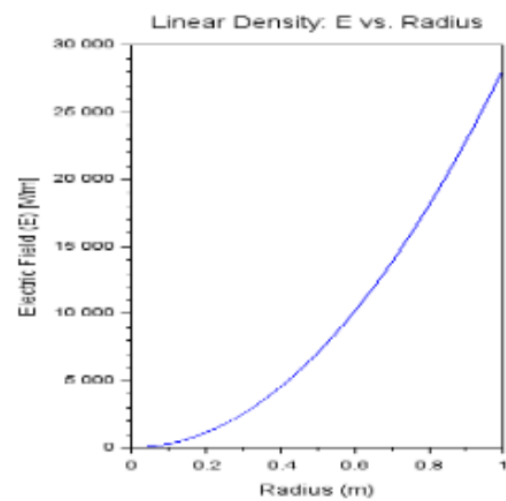
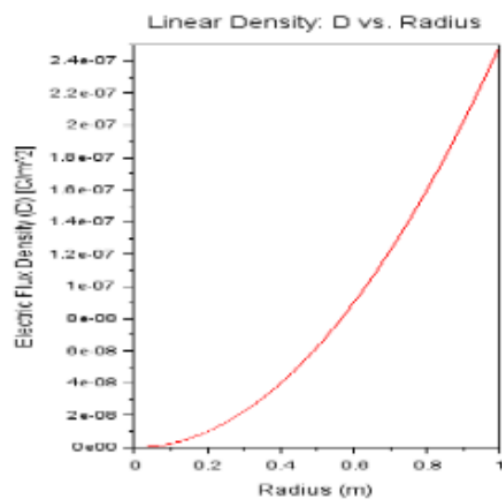
CASE 2:

PROGRAM:

```
EMF EX2 CASE 2.sc (C:\Users\rahu\OneDrive\Documents\EMF EX2 CASE 2.sc) - SciNotes
File Edit Format Options Window Execute ?
EMF EX2 CASE 2.sc (C:\Users\rahu\OneDrive\Documents\EMF EX2 CASE 2.sc) - SciNotes
EMF EX2 (I).sd X EMF EX2 (II).sd X EMF EX2 CASE 2.sd X EMF EXP2 CASE 3.sd X

5 n = 100; // Number of steps for plotting
6 r = linspace(0, R, n); // Radial distances
7 // Initialize arrays for D and E
8 D = zeros(1, n);
9 E = zeros(1, n);
10 // Calculate D and E
11 for i = 1:n
12     ri = r(i); // Current radius
13     // Enclosed charge Q_enclosed = rho0 * pi * ri^2
14     Q_enclosed = rho0 * pi * ri^2;
15
16     // Electric Flux Density D
17     if ri == 0 then
18         D(i) = Q_enclosed / (4 * pi * ri^2);
19     else
20         D(i) = 0; // Avoid division by zero
21     end
22
23     // Electric Field E = D / epsilon
24     E(i) = D(i) / epsilon;
25 end
26 // Plotting
27 subplot(1,2,1);
28 hold on;
29 xlabel('Radius (m)');
30 ylabel('Electric Flux Density (D) - [C/m^2]');
31 title('Linear Density: D vs. Radius');
32
33 subplot(1,2,2);
34 hold on;
35 xlabel('Radius (m)');
36 ylabel('Electric Field (E) - [V/m]');
37 title('Linear Density: E vs. Radius');
38
```

OUTPUT:



CASE 3:

PROGRAM:

```
EMF EXP2 CASE 3.sci (C:\Users\rahul\OneDrive\Documents\EMF EXP2 CASE 3.sci) - SciNotes
File Edit Format Options Window Execute ?
EMF EXP2 CASE 3.sci (C:\Users\rahul\OneDrive\Documents\EMF EXP2 CASE 3.sci) - SciNotes
EMF EXP2 (I).sd X EMF EXP2 (II).sd X EMF EXP2 CASE 2.sd X EMF EXP2 CASE 3.sd X

1 rho0=3e-6; //Uniform charge density (C/m^3)
2 epsilon = 8.854e-12; // Permittivity of free space (F/m)
3 R=2; // Radius of the sphere (m)
4 n = 200; // Number of steps for plotting
5 r = linspace(0, 4, n); // Radial distances from 0 to 4 meters
6 // Initialize arrays for D and E
7 D = zeros(1, n);
8 E = zeros(1, n);
9
10 // Calculate the total charge enclosed in the sphere
11 Q_total = rho0 * (4 * pi * R^3 / 3);
12
13 // Calculate D and E
14 for i = 1:n
15     ri = r(i); // Current radius
16
17     // Enclosed charge for r < R (inside the sphere)
18     if ri < R then
19         Q_enclosed = rho0 * (4 * pi * ri^3 / 3);
20     // Enclosed charge for r >= R (outside the sphere)
21     else
22         Q_enclosed = Q_total;
23     end
24
25     // Electric Flux Density D
26     D(i) = Q_enclosed / (4 * pi * ri^2);
27
28     // Electric Field E = D ./ epsilon
29     E(i) = D(i) / epsilon;
30 end
31 // Plotting
32 subplot(1, 2, 1);
33 plot(r, D, "r");
34 xlabel("Radius (m)");
```

OUTPUT:

